SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
School of Computing | Department of Data Science and Business Systems

UNIT 2 - TASK 1a: CREDIT RISK SCORECARD MODELING REPORT

Application Scorecard Modeling, WOE/IV Binning & Expected Loss (PD, LGD, EAD) Quantification

Course: 21CSE449T – Risk Analytics | Faculty: Dr. V. Prasanna | B.Tech – CSE (BDA & DS)

1. Model Discrimination & Performance Summary

Validation Metric	Train Set (N=750)	Test Set (N=250)	Industry Benchmark (Basel III)
ROC-AUC	0.8262	**0.7765**	>= 0.75
Gini Coefficient (2*AUC - 1)	0.6523	**0.5529**	>= 0.50
Kolmogorov-Smirnov (KS)	53.78%	**44.00%**	>= 40.0%

2. Portfolio Expected Loss (EL) Breakdown by Credit Rating Grade

Rating Grade	Count	Default %	Avg Score	Avg PD %	Total EAD (€)	Total Expected Loss (€)	EL % EAD
Grade F (High Risk / Subprime)	147	68.03%	681.8	72.43%	€766,077	**€233,613**	30.49%
Grade D (Vulnerable / Speculative)	229	50.66%	716.2	45.77%	€832,236	**€142,185**	17.08%
Grade C (Moderate Risk)	159	25.79%	740.7	26.59%	€476,081	**€47,823**	10.05%
Grade B (Acceptable / Investment Grade)	166	14.46%	760.7	15.41%	€462,474	**€25,005**	5.41%
Grade A (Prime / Low Risk)	162	9.26%	782.5	7.92%	€404,513	**€11,348**	2.81%
Grade AAA (Super Prime / Pristine)	137	2.92%	814.2	2.99%	€329,877	**€3,073**	0.93%



